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Student ID: 14340308

A UAV-Based Comparative Evaluation of Beaver Reintroduction Effects on Woodland Canopy Structure and Proxy-Based Understory Floristic Biodiversity at Woodland Valley Farm

Introduction

Beavers (*Castor* spp.) are known as ecosystem engineers because their tree felling, dam building, and water management activities can significantly alter woodland structure. These changes often influence how open the canopy is and how varied the habitat becomes, which in turn affects the diversity of plants growing beneath. However, the impact of beavers is not always the same—it depends on local conditions, the scale of the area, and how long beavers have been present. For this reason, each site needs to be assessed on its own terms rather than assuming their effects are universally positive (Stringer & Gaywood, 2016).

Understory vegetation dynamics are closely linked to canopy characteristics because canopy gaps control light penetration and opportunities for regeneration. The geometric properties of canopy gaps can therefore serve as a broad proxy for understory species diversity, especially in light-limited environments (Getzin et al., 2012). UAV-based remote sensing offers a way to map these gaps at high spatial detail, allowing comparative analysis where on the ground biodiversity surveys are not feasible (Bagaram et al., 2018).

This study compares two woodland areas at Woodland Valley Farm to explore how beaver activity might affect the diversity of plants growing beneath the canopy. Using UAV data, canopy gap structures were mapped and analysed for both sites, one influenced by beavers and one without. A gap shape complexity index and an empirical proxy for estimating understory species richness were then applied to see whether the structural differences matched what might be expected from beaver activity. Throughout the analysis, attention was given to uncertainty and the limits of what can reasonably be inferred from these data.

Methods

This study compared canopy gap patterns across two woodland areas at Woodland Valley Farm: one influenced by beaver activity (Woodland-1) and another without beaver presence (Woodland-2). To ensure a fair comparison, both sites were processed using the same workflow, parameter settings, and analysis thresholds. High resolution UAV images were collected during fieldwork at each site, with ground control points (GCPs) set up to keep the spatial data accurate and consistent. Structure-from-Motion (SfM) processing generated orthomosaics and digital surface models (DSMs), supplied in the module dataset and used as the basis for analysis. All spatial processing was carried out within a consistent projected coordinate system (EPSG:27700) to ensure reliable area and perimeter calculations.

Woodland boundaries were digitised manually from the orthomosaics to isolate canopy-covered areas and reduce potential edge effects. To represent canopy height, a digital terrain model (DTM) was interpolated from available ground points using inverse distance weighting and subtracted from the DSM to create a height-above-ground surface. Canopy gaps were identified by reclassifying this surface using a fixed 7 m height threshold, in line with the practical specification, and verified against the UAV orthomosaics. The resulting gap rasters were masked to woodland extents, converted to vector polygons, and clipped to keep the two sites distinct. For each canopy gap identified, both the area and perimeter were measured to calculate the Gap Shape Complexity Index (GSCI), which indicates how irregular the gap's shape is compared with a perfect circle of the same area.

Following the practical instructions, the analysis concentrated on the three largest canopy gaps in each woodland. The median GSCI from these was then used as a simple way to summarise the main structural differences between the two sites. Understory plant diversity wasn't measured directly; instead, predicted species richness was estimated using the provided empirical model, treating canopy gap structure as an indirect comparison tool. During the UAV survey, field observations in Woodland-1 recorded beaver dams, piles of woody debris, and areas of local flooding, providing useful context for the kinds of disturbance processes that may have shaped the canopy structure there.

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Figure A: Evident tree felling and localised flooding associated with beaver activity



Figure B: Beaver dam and woody debris accumulation



Results

Analysis of the UAV-derived canopy gap shows clear structural variation between the two woodland sites (Figure 1a–b). In both cases, canopy gaps form a mosaic of openings scattered throughout the woodland interior. However, the beaver-influenced woodland (Woodland-1) displays a dominant, spatially continuous opening, while the woodland without beavers (Woodland-2) shows a more fragmented pattern composed of several smaller gaps. This contrast is further supported by the distribution of the three largest gaps, which together occupy a larger share of the total gap area in Woodland-1 (Figure 2a–b).

Figure 1a: Canopy Gap Distribution (Woodland-1)

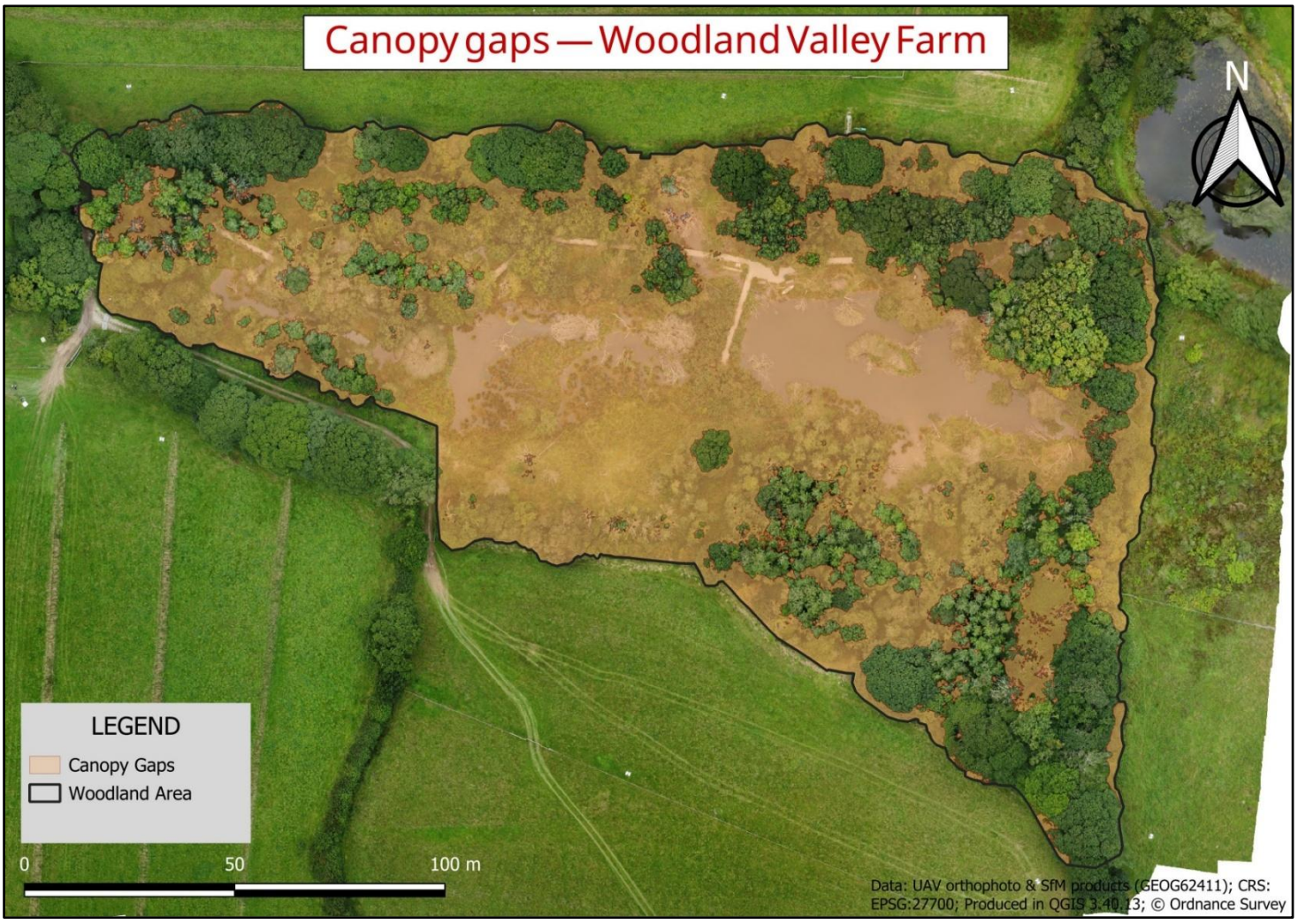
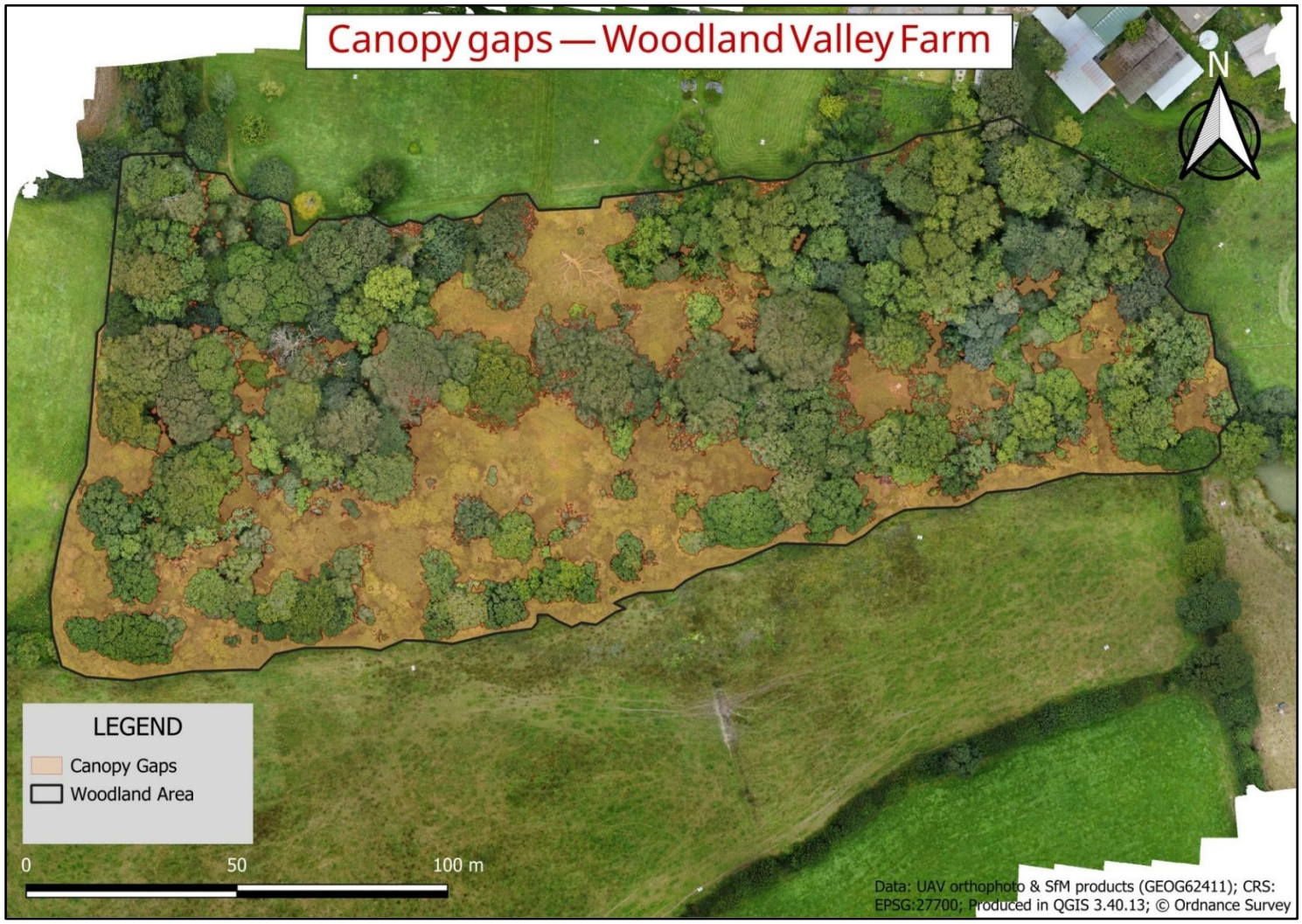


Figure 1b: Canopy Gap Distribution (Woodland-2)



The three largest canopy gaps in each woodland were used to evaluate gap shape complexity through the median Gap Shape Complexity Index (GSCI). Median GSCI was slightly lower in Woodland-1 (13.96) compared to Woodland-2 (15.08), suggesting that the dominant gaps in the beaver-influenced site had simpler shapes overall. Using the provided regression relationship, predicted understory species richness was estimated at roughly 203 species (±12) for Woodland-1 and 218 species (±12) for Woodland-2. Although Woodland-1 contained larger, more continuous openings, the proxy model therefore indicates a higher predicted richness in Woodland-2 by around 16 species, revealing a divergence between structural dominance and the proxy-derived estimates of biodiversity.

Figure 2a: Top 3 Gaps (Woodland-1)

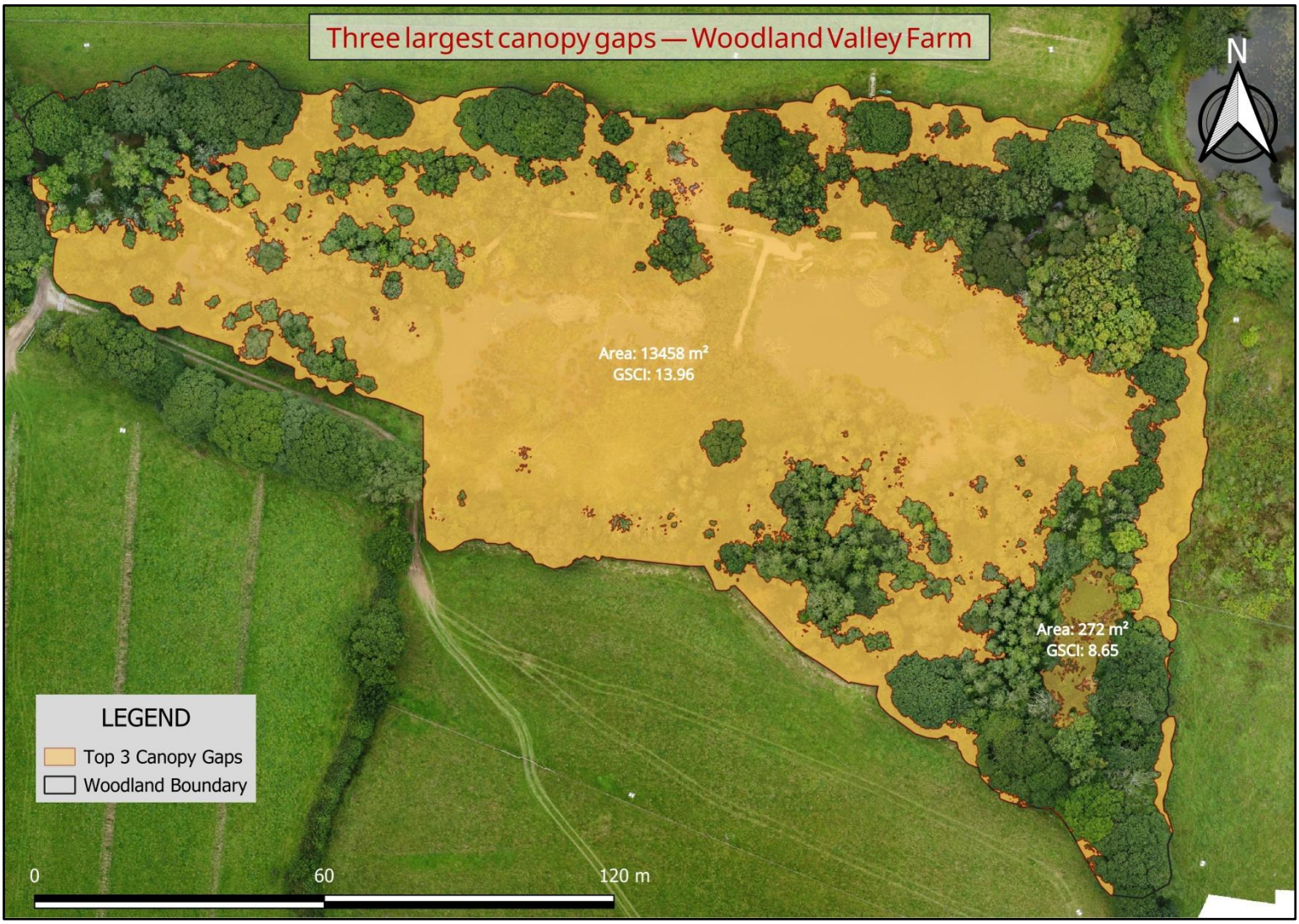
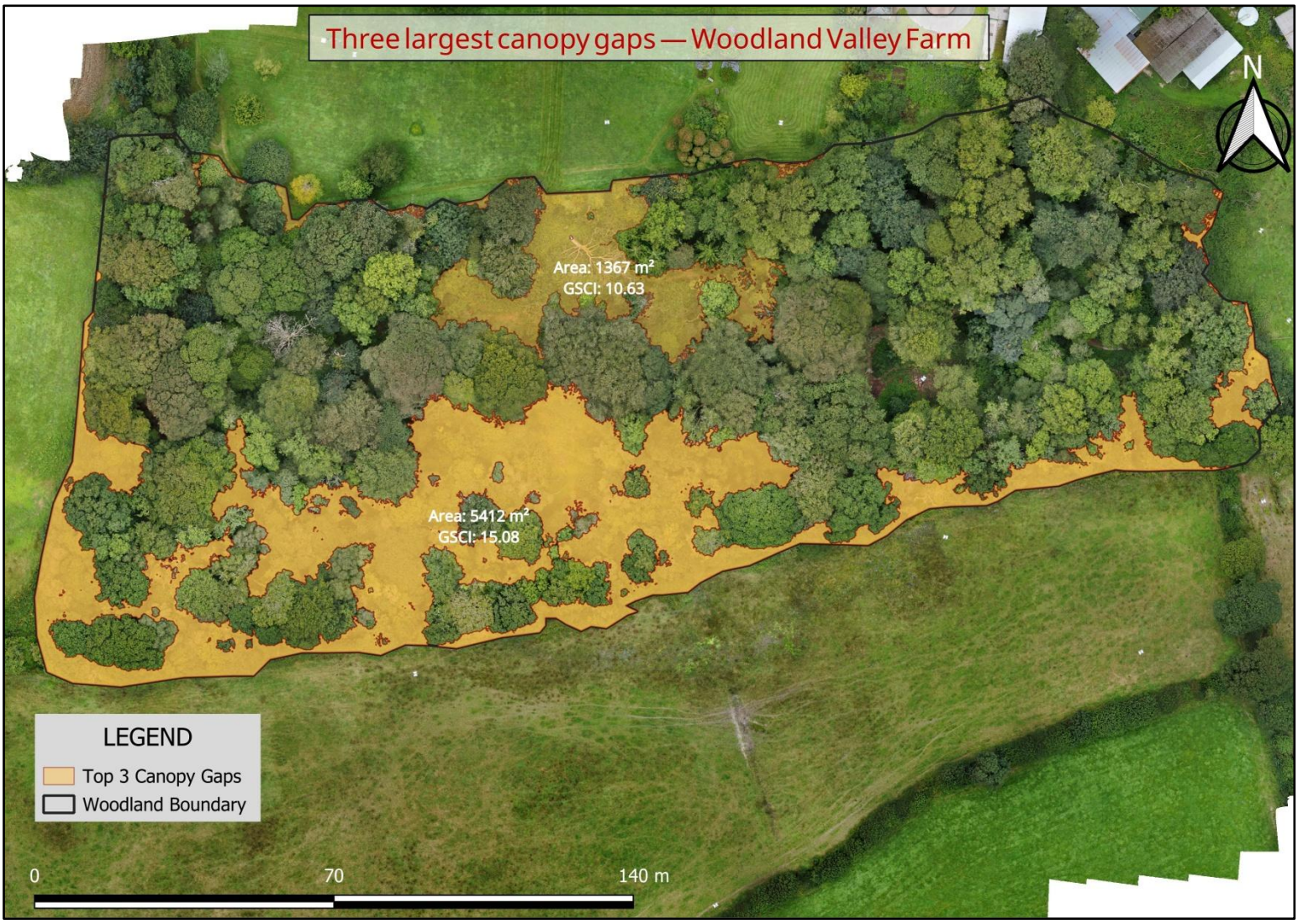


Figure 2b: Top 3 Gaps (Woodland-2)



Discussion

This study set out to examine whether differences in canopy gap structure between two woodlands at Woodland Valley Farm reflect the expected ecological effects of beaver activity. UAV-derived gap geometry was used as an indirect way to assess understory floristic biodiversity, and clear structural differences were identified between the two sites. The beaver-influenced woodland (Woodland-1) had fewer canopy gaps overall, but they were larger and more continuous. In contrast, the woodland without beavers (Woodland-2) showed a more broken-up structure, made up of many smaller gaps that were more evenly spread in size. This suggests differences not just in how gaps are distributed, but also in the kind of disturbance shaping the canopy. Woodland-1 seems to reflect processes that create broader, longer-lasting openings, whereas Woodland-2 shows a pattern more typical of smaller, gradual canopy loss.

Within the proxy framework used in this study, the structural differences observed could have implications for understory biodiversity. Larger and more irregular canopy gaps tend to let in more light, create a broader range of microclimates, and offer more opportunities for regeneration, which can support higher species richness than smaller, fragmented gaps. This idea forms the basis for using the median Gap Shape Complexity Index (GSCI) from the three largest gaps, as it captures the ecological significance of the main openings rather than the combined influence of numerous smaller ones. Previous fine-scale ecological research has shown that gap-based structural metrics can explain considerable variation in understory plant richness in systems where light limits regeneration (Getzin et al., 2012). However, in this study, biodiversity values derived from GSCI reflect relative, modelled estimates rather than measured floristic data, so they should be understood in comparative rather than absolute terms.

Beaver activity offers a credible ecological mechanism to explain the canopy pattern found in Woodland-1, although causation cannot be confirmed directly. Beavers act as ecosystem engineers through tree felling, dam construction, and creating localised flooding, often leading to large canopy openings, standing deadwood, and lingering structural disturbance (Stringer & Gaywood, 2016). The pattern seen in Woodland-1, where a few large gaps dominate, fits with the kind of spatially clustered and episodic disturbance typically linked to beaver activity, rather than gradual or human-managed canopy removal. Field observations and previous research suggest that beaver tree-felling can significantly increase canopy openness and alter conditions on the forest floor, especially close to water bodies and depending on how recently and intensively the beavers have been active (Zwolicki, 2025). Even so, because the present analysis did not directly map signs of beaver felling, dam positions, or disturbance chronology, beaver influence should be regarded as a likely contributing factor rather than a confirmed cause.

A number of methodological, ecological, and interpretive uncertainties limit how strongly conclusions can be drawn. From a methodological side, canopy gaps were identified using a fixed 7 m height threshold derived from the UAV data, means that the results could be influenced by factors like lighting conditions, shadows, and the way gaps were delineated—particularly in smaller or partially shaded areas (Bagaram et al., 2018). By focusing only on the three largest gaps, the analysis also becomes more sensitive to small differences in how boundaries were drawn; even minor adjustments could change the median GSCI values and the associated species-richness estimates. Ecologically, the proxy linking canopy structure to

understory richness was originally developed in a different woodland environment, so there's some uncertainty about how well it fits the conditions at Woodland Valley Farm (Getzin et al., 2012). Finally, because the study only compares two sites, the results can't be generalised beyond this specific context. These findings are best understood as site-specific indications rather than broad or causal conclusions. Collectively, the evidence points to structural differences likely associated with beaver presence, but it shouldn't be interpreted as proof of actual changes in biodiversity.

Overall, the results show that UAV-derived canopy gap metrics can clearly distinguish structural differences between the beaver-influenced and non-beaver woodlands at Woodland Valley Farm. Although the proxy estimates of biodiversity don't directly match the dominance of larger gaps, the patterns observed offer a helpful starting point for understanding how beaver-related structural changes might shape ecological conditions under different canopy arrangements.

Conclusion

This study used UAV-derived canopy gap analysis to explore how beaver reintroduction might influence woodland understory biodiversity at Woodland Valley Farm. Noticeable differences in gap layout and shape complexity were found between the beaver and non-beaver sites, leading to contrasting proxy estimates of understory species richness. While these patterns align with current understanding of how beavers can affect woodland structure, the conclusions rest on an indirect proxy and are shaped by both methodological and ecological uncertainty. Overall, the findings should be viewed as indicative rather than conclusive, pointing to possible structural effects associated with beaver activity and highlighting how UAV-based analysis can support careful, site-specific ecological research.

References

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